

SHENANDOAH VALLEY, VA, USA

WMO: 724105

Lat: 38.264N

Lon: 78.896W

Elev: 1201

StdP: 14.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93760

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	16.0	-4.4	4.6	17.6	0.9	6.0	22.8	20.4	42.3	18.5	42.6	2.7	310	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	93.0	73.8	90.6	73.3	88.3	72.5	77.9	87.0	76.4	85.6	75.0	84.0	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	138.0	82.3	73.3	129.4	80.3	72.4	125.6	79.4	42.2	86.9	40.7	86.2	39.4	83.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	15.3	12.6		2.1	96.6	6.7	3.1	-2.8	98.9	-6.7	100.7	-10.5	102.5	-15.4	104.7
				2.2	79.5	5.3	2.6	-1.6	81.3	-4.7	82.8	-7.6	84.3	-11.5	86.2
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	35.2	37.8	45.1	55.3	64.3	72.3	75.8	74.4	67.6	56.4	45.5	38.3	(6)
	DBStd	16.61	10.96	10.21	9.91	8.48	7.44	5.69	4.63	4.98	6.51	8.12	8.82	9.40	(7)
	HDD50	1694	477	356	217	40	2	0	0	0	0	30	190	382	(8)
	HDD65	4480	925	761	620	308	106	10	1	2	48	285	586	828	(9)
	CDD50	3800	17	16	64	198	446	670	801	757	529	229	54	19	(10)
	CDD65	1113	0	0	2	16	86	230	337	294	127	20	1	0	(11)
	CDH74	10834	1	1	43	305	922	2196	3221	2692	1177	254	19	3	(12)
	CDH80	4117	0	0	5	74	272	843	1379	1087	399	56	2	0	(13)
(14) Wind	WSAvg	4.1	5.0	5.0	5.4	5.2	4.2	3.7	3.1	2.8	3.2	3.5	3.9	3.9	(14)
(15) Precipitation	PrecAvg	39.8	2.7	2.4	3.7	3.1	3.7	3.7	4.1	3.7	4.3	2.7	2.7	3.1	(15)
	PrecMax	60.1	7.5	6.7	7.0	4.9	6.4	8.1	7.9	6.9	12.9	8.9	5.1	5.4	(16)
	PrecMin	29.3	0.8	0.5	0.8	1.1	1.4	0.6	0.9	1.8	0.4	0.0	0.5	1.8	(17)
	PrecStd	7.7	1.9	1.5	1.5	1.2	1.6	2.2	1.5	1.4	3.3	2.2	1.4	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	70.9	79.4	87.7	90.2	94.6	97.5	95.5	91.5	85.1	76.5	71.7	(19)
		MCWB	59.6	57.7	61.1	65.8	70.6	73.7	74.0	75.9	71.6	69.6	62.4	61.3	(20)
	2%	DB	63.7	64.3	73.0	81.6	86.2	90.9	93.1	92.1	88.4	81.0	71.2	63.9	(21)
		MCWB	56.0	52.6	57.8	62.9	69.4	73.1	74.0	74.8	70.7	66.4	58.9	55.8	(22)
	5%	DB	57.8	60.5	68.1	77.3	82.5	88.3	90.5	89.9	84.3	75.6	66.0	59.1	(23)
		MCWB	51.7	50.5	55.3	60.9	67.7	72.5	73.7	73.8	69.7	63.8	57.3	52.8	(24)
	10%	DB	52.7	54.8	63.4	72.5	79.5	84.7	87.9	86.2	81.2	72.1	62.5	54.4	(25)
		MCWB	46.4	46.3	52.6	58.6	66.1	70.7	73.4	72.4	68.6	61.9	54.3	48.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	60.1	63.8	69.4	74.2	78.0	79.5	81.2	76.8	73.1	66.3	63.6	(27)
		MCDB	66.2	67.7	74.7	81.1	84.7	86.7	90.0	89.1	84.7	81.8	72.2	68.5	(28)
	2%	WB	58.2	55.7	60.4	65.9	71.8	76.1	77.5	78.4	74.3	70.1	62.7	57.9	(29)
		MCDB	63.1	61.7	69.8	76.4	81.7	86.0	87.3	86.9	82.3	76.3	68.2	63.4	(30)
	5%	WB	52.7	51.5	57.3	63.5	70.1	74.6	76.1	76.6	72.5	66.9	59.0	53.6	(31)
		MCDB	57.2	57.7	65.5	72.4	79.9	84.3	85.8	85.1	80.1	73.2	64.6	57.6	(32)
	10%	WB	46.6	47.1	53.7	60.9	68.2	72.9	74.7	74.8	70.7	63.9	55.1	48.6	(33)
		MCDB	51.2	54.1	61.4	70.1	76.6	81.9	84.0	82.9	77.9	69.7	60.9	53.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	20.8	22.7	24.9	23.1	21.9	21.6	21.7	22.5	23.8	22.8	19.8	(35)
		MCDBR	24.7	28.7	30.5	32.2	27.7	25.9	25.5	25.9	28.2	29.9	29.7	26.1	(36)
		MCWBR	18.1	19.3	17.7	16.5	13.5	11.0	9.7	10.5	12.9	16.5	19.6	18.7	(37)
	5% WB	MCDBR	22.6	24.2	26.8	26.3	24.4	23.0	21.7	21.9	21.5	24.0	24.1	22.8	(38)
		MCWBR	18.4	18.6	17.0	14.9	12.7	10.8	9.6	10.5	11.1	14.7	18.3	18.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.294	0.315	0.346	0.391	0.430	0.478	0.492	0.505	0.394	0.350	0.315	0.295	(40)
	taud		2.499	2.422	2.394	2.293	2.203	2.104	2.091	2.025	2.331	2.447	2.505	2.525	(41)
	Ebn at Noon		284	290	290	281	271	257	252	245	270	273	272	275	(42)
	Edh at Noon		25	30	35	41	46	50	50	53	37	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		685	938	1254	1573	1730	1907	1825	1653	1375	1066	780	602	(44)
	RadStd		57	112	75	122	148	124	108	80	162	122	64	50	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		-0.79	N/A	N/A	-1.76	-1.57	-1.76	N/A	N/A	N/A	-128
(47) Regional (43 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+230	+131

Nomenclature: See separate page